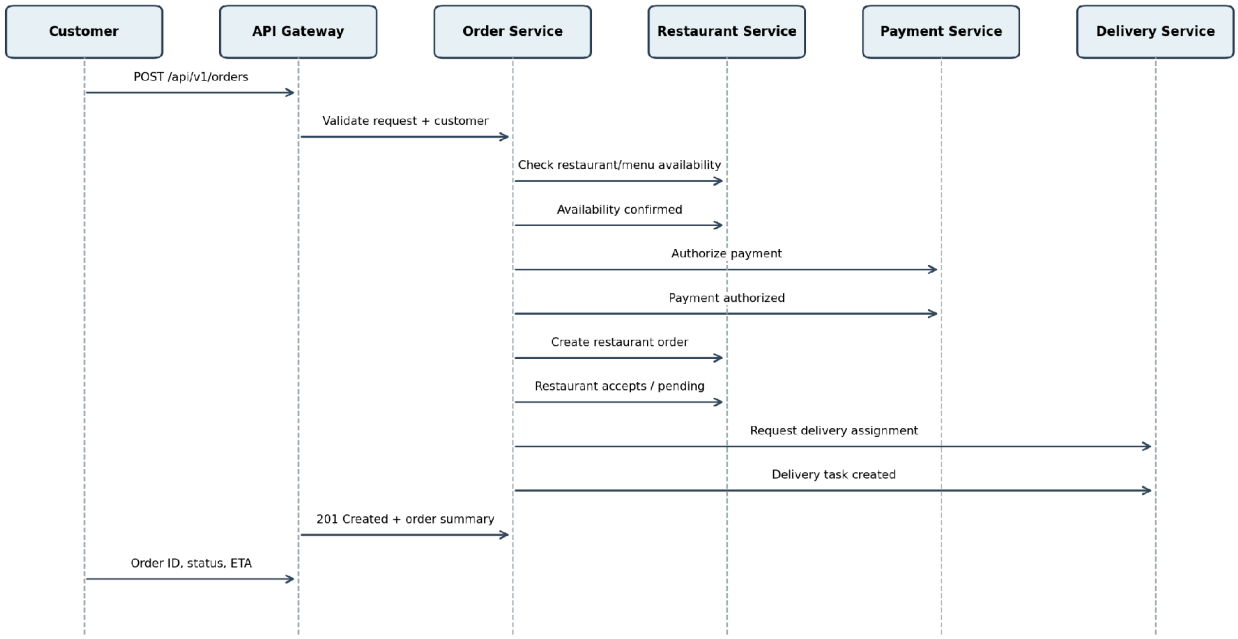
ASSIGNMENT 5

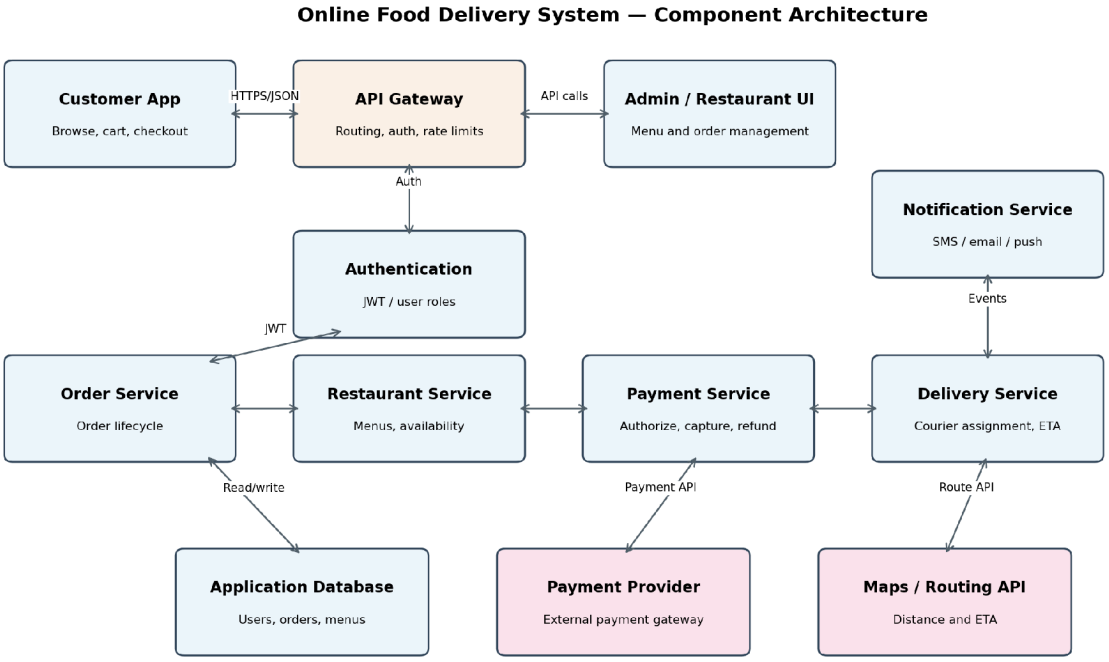
1. Sequence Diagram



Typical successful order-placement flow (simplified)

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2. Component Diagram



3. API Specification Document

Base URL: https://api.fooddelivery.example/api/v1 (illustrative). Protected endpoints require Authorization: Bearer <access-token>. Content-Type: application/json.

3.1 Endpoint Catalogue

Service	Method + Endpoint	Auth	Request	Success response
Authentication	POST /auth/register	Public	Request: {"name":"Asha Patil","email":"asha@example.com","password":"StrongPass123!","phone":"+919876543210"}	201 Created: {"userId":"uuid","name":"Asha Patil","email":"asha@example.com"}
Authentication	POST /auth/login	Public	Request: {"email":"asha@example.com","password":"StrongPass123!"}	200 OK: {"accessToken":"jwt-token","tokenType":"Bearer","expiresIn":3600,"user":{"id":"uuid","name":"Asha Patil"}}
Restaurants	GET /restaurants	Public	Query: ?city=Pune&cuisine=Indian&page=1&limit=20	200 OK: {"items":[{"id":"uuid","name":"Spice House","cuisine":["Indian"],"rating":4.4,"isOpen":true}],"page":1,"limit":20,"total":1}
Restaurants	GET /restaurants/{restaurantId}	Public	Path: restaurantId (UUID)	200 OK: {"id":"uuid","name":"Spice House","address":"Pune","isOpen":true,"estimatedDeliveryMinutes":35}
Menu	GET /restaurants/{restaurantId}/menu	Public	Path: restaurantId; optional query ?category=Main Course	200 OK: {"restaurantId":"uuid","items":[{"id":"item-uuid","name":"Paneer Thali","description":"Meal","price":220.00,"currency":"INR","available":true}]}
Cart / Orders	POST /orders	Customer JWT	Request: {"restaurantId":"uuid","items":[{"menuItemId":"item-uuid","quantity":2}],"deliveryAddress":{"line1":"12 MG Road","city":"Pune","postalCode":"411001","latitude":18.52,"longitude":73.85},"paymentMethodId":"pm-123","specialInstructions":"Less spicy"}	201 Created: {"orderId":"uuid","status":"PLACED","subtotal":440.00,"deliveryFee":30.00,"tax":22.00,"total":492.00,"currency":"INR","estimatedDeliveryAt":"2026-10-15T18:45:00Z"}
Orders	GET /orders/{orderId}	Customer JWT; owner or authorize	Path: orderId (UUID)	200 OK: {"orderId":"uuid","status":"PREPARING","items":[{"menuItemId":"item-uuid","name":"Paneer Thali","quantity":2,"unitPrice":220.00}],"total":492.00,"currency":"INR","createdAt":"2026-10-15T18:00:00Z"}

		d staff		
Orders	GET /orders	Customer JWT	Query: ?status=DELIVERED&page=1&limit=10	200 OK: {"items":[{"orderId":"uuid","status":"DELIVERED","total":492.00,"currency":"INR"}],"page":1,"limit":10,"total":1}
Orders	POST /orders/{orderId}/cancel	Customer JWT; cancellation policy applies	Optional body: {"reason":"Ordered by mistake"}	200 OK: {"orderId":"uuid","status":"CANCELLED","refundStatus":"PENDING"}
Restaurant Orders	PATCH /restaurant/orders/{orderId}/status	Restaurant staff JWT	Request: {"status":"ACCEPTED","preparationMinutes":25}	200 OK: {"orderId":"uuid","status":"ACCEPTED","estimatedReadyAt":"2026-10-15T18:25:00Z"}
Payments	GET /orders/{orderId}/payment	Customer JWT; owner or authorized staff	Path: orderId (UUID)	200 OK: {"paymentId":"uuid","orderId":"uuid","status":"AUTHORIZED","amount":492.00,"currency":"INR","providerReference":"pay_abc123"}
Delivery	GET /orders/{orderId}/tracking	Customer JWT; owner or authorized staff	Path: orderId (UUID)	200 OK: {"orderId":"uuid","deliveryStatus":"OUT_FOR_DELIVERY","courier":{"name":"Ravi","phoneMasked":"******3210"},"currentLocation":{"latitude":18.53,"longitude":73.84},"estimatedDeliveryAt":"2026-10-15T18:45:00Z","updatedAt":"2026-10-15T18:20:00Z"}
Menu Management	POST /restaurant/menu-items	Restaurant staff JWT	Request: {"name":"Paneer Thali","description":"Meal","price":220.00,"currency":"INR","category":"Main Course","available":true}	201 Created: {"id":"item-uuid","name":"Paneer Thali","price":220.00,"currency":"INR","available":true}
Menu Management	PUT /restaurant/menu-items/{itemId}	Restaurant staff JWT	Request: {"name":"Paneer Special Thali","description":"Meal","price":250.00,"currency":"INR","category":"Main Course","available":true}	200 OK: {"id":"item-uuid","name":"Paneer Special Thali","price":250.00,"currency":"INR","available":true}
Menu Management	DELETE /restaurant/menu-items/{itemId}	Restaurant staff JWT	Path: itemId (UUID); soft-delete recommended if referenced by orders	204 No Content

3.2 Common Error Response

All errors use a consistent JSON structure. Example: {"error":{"code":"ORDER_STATE_CONFLICT","message":"This order cannot be cancelled in its current state.","requestId":"req-123","details":[]}}

3.3 HTTP Status Codes

200 OK: request succeeded. 201 Created: resource created. 204 No Content: successful deletion with no response body. 400 Bad Request: malformed request. 401 Unauthorized: authentication required/invalid. 403 Forbidden: insufficient permissions. 404 Not Found: resource absent. 409 Conflict: current state prevents operation. 422 Unprocessable Entity: business validation failure. 429 Too Many Requests: throttled. 500 Internal Server Error: unexpected failure. 502/503: upstream or temporary service failure.

3.4 Endpoint-by-Endpoint Error Codes

POST /auth/register — 400, 409, 422

POST /auth/login — 400, 401, 422

GET /restaurants — 400

GET /restaurants/{restaurantId} — 404

GET /restaurants/{restaurantId}/menu — 404

POST /orders — 400, 401, 404, 409, 422, 402

GET /orders/{orderId} — 401, 403, 404

GET /orders — 400, 401

POST /orders/{orderId}/cancel — 401, 403, 404, 409, 422

PATCH /restaurant/orders/{orderId}/status — 400, 401, 403, 404, 409, 422

GET /orders/{orderId}/payment — 401, 403, 404

GET /orders/{orderId}/tracking — 401, 403, 404, 409

POST /restaurant/menu-items — 400, 401, 403, 422

PUT /restaurant/menu-items/{itemId} — 400, 401, 403, 404, 422

DELETE /restaurant/menu-items/{itemId} — 401, 403, 404, 409

4. Non-Functional and Security Considerations

- Use HTTPS for all external traffic and validate all request bodies against schemas.
- Apply role-based access control: customers access only their own orders; restaurant staff update only their restaurant's orders and menu.
- Calculate totals and taxes on the server; never trust client-supplied prices or totals.
- Use idempotency keys for order creation and payment operations to prevent duplicate charges/orders.
- Do not store raw card numbers or CVV. Use tokenized payment-provider references.
- Log request IDs, state transitions, and provider errors; do not log passwords, access tokens, or payment secrets.
- Use pagination for collection endpoints, rate limits for login/checkout, and retries with backoff for transient upstream failures.
- Publish order state changes as events (for example, OrderPlaced, PaymentAuthorized, OrderAccepted) to decouple notifications and delivery assignment.